Inferential data Analysis

	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio	Selector
0	65	Female	0.7	0.10	187.00	16.0	18	6.8	3.3	0.90	1
1	62	Male	5.3	2.95	481.75	64.0	100	7.5	3.2	0.74	1
2	62	Male	5.3	2.95	481.75	60.0	68	7.0	3.3	0.89	1
3	58	Male	1.0	0.40	182.00	14.0	20	6.8	3.4	1.00	1
4	72	Male	3.9	2.00	195.00	27.0	59	7.3	2.4	0.40	1
...	...	...	...	...	...	...	...	...	...	...	...
578	60	Male	0.5	0.10	481.75	20.0	34	5.9	1.6	0.37	0
579	40	Male	0.6	0.10	98.00	35.0	31	6.0	3.2	1.10	1
580	52	Male	0.8	0.20	245.00	48.0	49	6.4	3.2	1.00	1
581	31	Male	1.3	0.50	184.00	29.0	32	6.8	3.4	1.00	1
582	38	Male	1.0	0.30	216.00	21.0	24	7.3	4.4	1.50	0

583 rows × 11 columns

Covariance: Difference between the columns

- The feature 'TB' is positively covariance in 'DB' and 'Selector' features.
- The feature DB is positively covariance in sgot and selector (linear upward direction) and Negatively covariance (linear downward) in ALB features.
- The ALB is negatively covariance in TB, DB and Selector features.

Correlation: Correlative between the columns

- The feature ALB and Alkphos are correlated each other.

Ttest: Ttest is the test to similarity between given variables based on Mean.

- Unpaired ttest-Independent variable-different groups with same condition.
- Paired ttest-Dependent variable-same group and different condition.
- Through pvalue can find the similarity. If $p<0.05$, Reject null hypothesis else $p>=0.05$,Fail to reject .
- H0(null hypothesis)-There is no similarity between given variables
- H1-There is a similarity difference between given variables

Indep_ttest

```
indep_ttest(dataset,'Gender','Male','Female','TP')
```

```
P_Value: 0.0321835159068134
Reject null hypothesis:There is a significant relationship between given varables
TtestResult(statistic=np.float64(-2.1472872216948455), pvalue=np.float64(0.0321835159068134), df=np.float64(581.0))
```

Dep_ttest

```
dep_ttest(dataset,'Gender','Male','Age','Selector')
```

```
P_Value: 1.8679108632189356e-207
Reject null hypothesis:There is a significant relationship between given varables
TtestResult(statistic=np.float64(57.74031154568498), pvalue=np.float64(1.8679108632189356e-207), df=np.int64(440))
```

Anova: Analysis of Variance

- Analysis of variance tells the significant difference between two variables
- H0-There is no significant difference between given variables
- H1-There is a significant difference between given variables

One_way Anova:

```
anova_oneway(dataset,'Sgpt','Sgot','TP')
```

```
one_way_pvalue: 2.7792482215077357e-142
Reject null hypothesis:There is a significant relationship between given varables
```

```
F_onewayResult(statistic=np.float64(395.128429098168), pvalue=np.float64(2.7792482215077357e-142))
```

```
#H0-fail to reject null hypothesis-There is no significant difference between given variables.
#H1-Reject null hypothesis-There is a significant difference between given variables.
#pvalue=0>0.05, if pvalue<0.05-Reject null hypothesis.
#So, here significant difference between Sgpt,Sgot,TP variables
```

Two_way Anova:

	df	sum_sq	mean_sq	F	PR(>F)
C(Gender)	1.0	488.011886	488.011886	1.901731	0.168417
C(Selector)	1.0	2704.198702	2704.198702	10.537979	0.001237
(Gender):C(Selector)	1.0	776.399129	776.399129	3.025546	0.082494
Residual	579.0	148579.819100	256.614541	NaN	NaN

```
Age + Gender=pvalue of 0.16>0.05 so H0 there is no significant difference between these two varables
Age + Selector=pvalue of 0<0.05 so H0 there is significant difference between these two varables
Gender:Selector=pvalue of 0.08>0.05 so H0 there is no significant difference between these two varabl
```

1.How many male and female affect liver disorder?

```
Gender
Female     92
Male      324
Name: Selector, dtype: int64
```

2. Who are all above 60 age affected person?

	Age	Gender	count
0	False	Male	361
1	False	Female	123
2	True	Male	80
3	True	Female	19

3. Reason for affected person?

	Age	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio	Selector
Selector										
0	41.239521	1.118563	0.390719	210.258982	32.784431	39.754491	6.543114	3.344311	1.024995	0.0
1	46.153846	2.243750	1.075962	267.717548	53.958534	74.901442	6.464904	3.060577	0.907318	1.0

When compared to person who has no disease, The affected person has high average value of the biochemical parameters. So it might be the reason for affected person.